

A COMPREHENSIVE LITERATURE REVIEW

Next-Gen KGQA RAG Systems

A rigorous structural analysis of modern differentiable, cognitive, goal-oriented, and multimodal retrieval-augmented generation paradigms over structured knowledge bases.

ACADEMIC PRESENTATION & SYNTHESIS REPORT

RAG Comparative Evaluation Matrix

Framework & Paper Reference	KGQA Methodology & Logic	Datasets & Benchmarks	Empirical Advancements	Gaps & Critical Bottlenecks
D-RAG (Gao et al., 2025) <i>2025.emnlp-main.1793.pdf</i>	Gumbel-Softmax differentiable reparameterization; GNN fact-level sampling.	WebQSP, CWQ, Freebase KG	WebQSP Hits@1: 89.1% (+2.5%), CWQ F1: 70.3% (+4.4%)	Relies on external entity linking; high GPU gradients cost.
KG-RAG (Wang et al., 2025) <i>s41598-025-21222-z.pdf</i>	Dual-Channel (DPR context + GNN path) with Path Attention.	Natural Questions, PubMedQA, UMLS	PubMedQA Accuracy: 81.3% (+6.8%)	Lacks standard dynamic update mechanisms.
Goal-Oriented (Yano et al., 2026) <i>information-17-00055-v3.pdf</i>	SPARQL translation from language; systemic process function trees.	Japan FSA Asset Audit	Decomposition F1: 0.83 vs. ChatGPT 4o (0.15)	Very high manual modeling cost by domain experts.
CogGRAG (Cheng et al., AAAI-26) <i>08507-AAAI26.ChengY-NLP.pdf</i>	Top-down tree-structured mind maps & dual-process verifier.	HotpotQA, CWQ, WebQSP, GRBench	GRBench-E-Commerce RL: 30.2% (+3.8%)	High API latency and recursive loop bottlenecks.
M³KG-RAG (Park et al., CVPR 2026) <i>Park_M3KG-RAG_..._paper.pdf</i>	Multi-agent MMKG construction & visual-audio grounding (GRASP).	AudioCaps-QA, VALOR, VCGPT	VALOR Qwen2.5 M.J. Score: 44.67 (+9.23)	Spatio-temporal graphs suffer processing lag.
KG²RAG (Zhu et al., 2025) <i>22.pdf</i>	Graph BFS triplet expansion & Maximum Spanning Tree context organizer.	HotpotQA, MuSiQue, TriviaQA	HotpotQA Retrieval F1: 0.436 (+0.09)	Extremely sensitive to graph connection density.
SG-RAG (Saleh et al., ICNLSP 2024) <i>2024.icnlsp-1.45.pdf</i>	Zero-shot Cypher template-based declarative subgraph retrieval.	MetaQA (1, 2, 3-Hop NTM)	MetaQA 3-Hop Hits: 58.0% vs. LLAMA3-8B (17.0%)	Rigid declarative structures; schema-translation dependent.
HyKGE (Jiang et al., ACL 2025) <i>2025.acl-long.580.pdf</i>	Hypothesis Outputs (HO) ELBO & fragment-level reranking.	MMCU-Med, CMB-Exam, CMB-Clin	MMCU-Medical EM: 57.82% (+14.3%)	Highly dependent on LLM zero-shot capability.

D-RAG: Differentiable Selection

Gao et al., 2025 | [2025.emnlp-main.1793.pdf](#)

Methodological Steps:

📦 **Fact Representation:** Concatenates GNN entity & relation embeddings
 $F_i = [h_i \parallel r_i \parallel t_i]$ to calculate selection probabilities $p(\tau_i) = \sigma(WF_i + b)$.

✂️ **Gumbel-Softmax Sampling:** Converts discrete selection into differentiable relaxation:

$$z_i^{\text{soft}} = \text{softmax} \left(\frac{[\log p_i + g_{i1}, \log(1 - p_i) + g_{i2}]}{t} \right)$$

🖋️ **Neural Prompting:** Enriches text embeddings with GNN structural representations to let gradient flow from LLM to GNN parameters.

Datasets & Gaps:

Datasets: WebQSP (62% 1-hop, 37% 2-hop) and CWQ (3-4 hops complex queries).

⚠️ **Critical Gap:** Fails when entity linking has upstream noise; high GPU gradient computational overhead.

 D-RAG Gradient flow concept mapping

KG-RAG: Dual-Channel Attention

Wang et al., 2025 | [s41598-025-21222-z.pdf](#)

Methodological Steps:

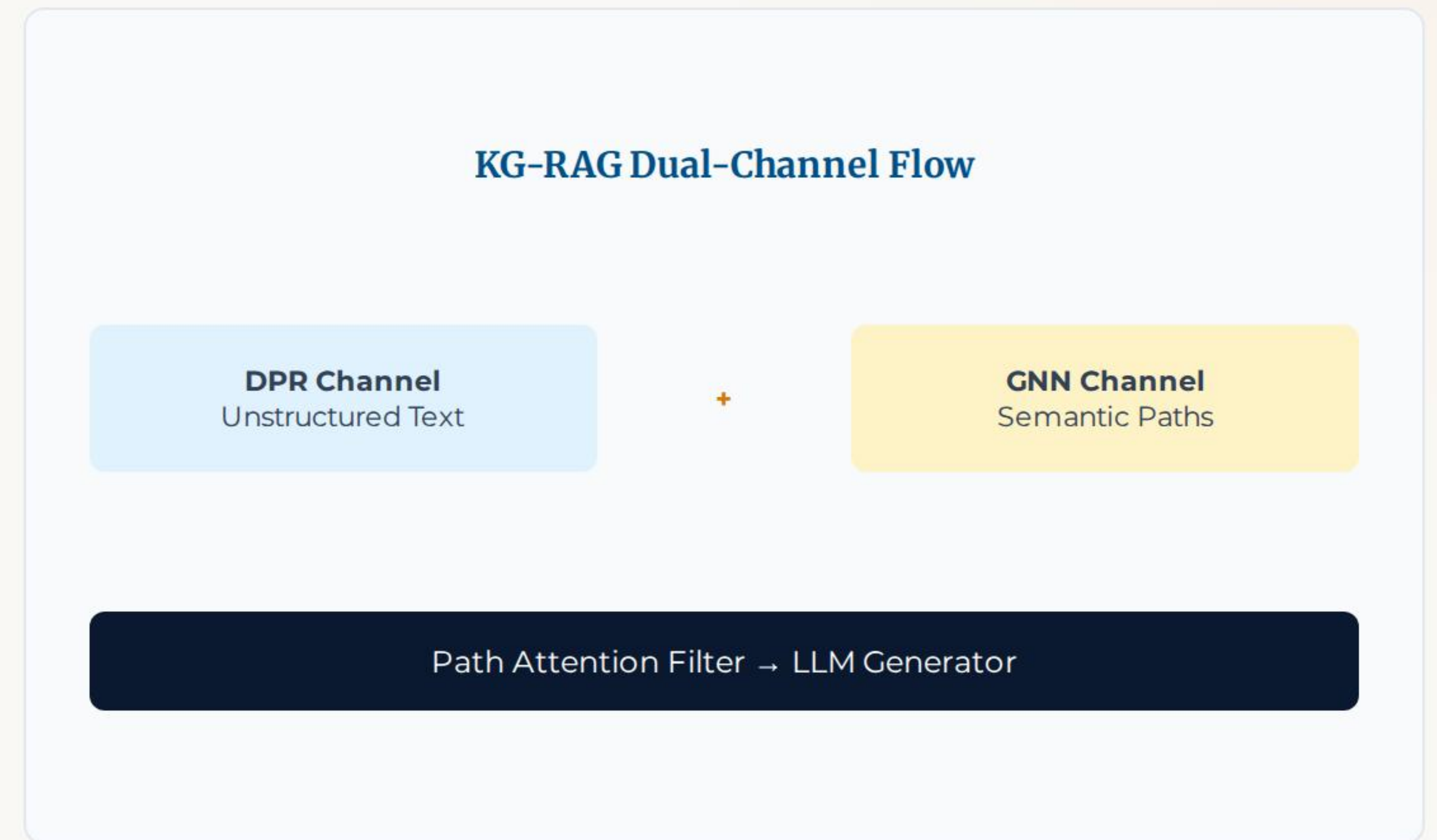
-  **Dual-Channel Retrieval:** Queries both unstructured text (DPR) and structured graphs (GNN path embeddings).
-  **Path Attention Scoring:** Assesses structural distance and similarity weights to align KG paths with query intent.
-  **Prompt Fusion:** Concatenates textual paragraphs and naturalized triplets in a hierarchical format.

Key Datasets & Gaps:

Datasets: Natural Questions (NQ) and PubMedQA (medical QA).

 ROUGE-L on NQ: 41.2 → **46.9** (+5.7), BLEU: 31.5 → **38.7**, PubMedQA Acc: **81.3%**.

 **Critical Gap:** Lacks standard mechanisms for dynamic graph updates, causing temporal staleness.



Goal-Oriented Action Decoupling

Yano et al., 2026 | *information-17-00055-v3.pdf*

Systemic Process trees:

Models human and system action states. Decouples actions into ****intended goal processes**** (actions/functions) and ****unintended goal processes**** (counter-functions).

</> Natural Language to SPARQL: Translates user commands into structured graph queries using few-shot prompt mappings.

@ Dependency Resolution: Models exclusive relationships and interlocking selections among way nodes.

Audit Metrics & Critical Gap:

Datasets: Japan FSA Asset Accounting Audit Standard Procedures (247 highly complex, interconnected nodes).

✓ Strict Precision: **0.83 F1 Score**. Significantly outperforms Microsoft GraphRAG (0.40) and vanilla ChatGPT (0.15).

⚠ **Critical Gap:** High modeling overhead. Trees must be manually engineered by domain experts, restricting generalizability.

CogGRAG: Human Mind Maps

Cheng et al., AAAI-26 | [08507-AAAI26.ChengY-NLP.pdf](#)

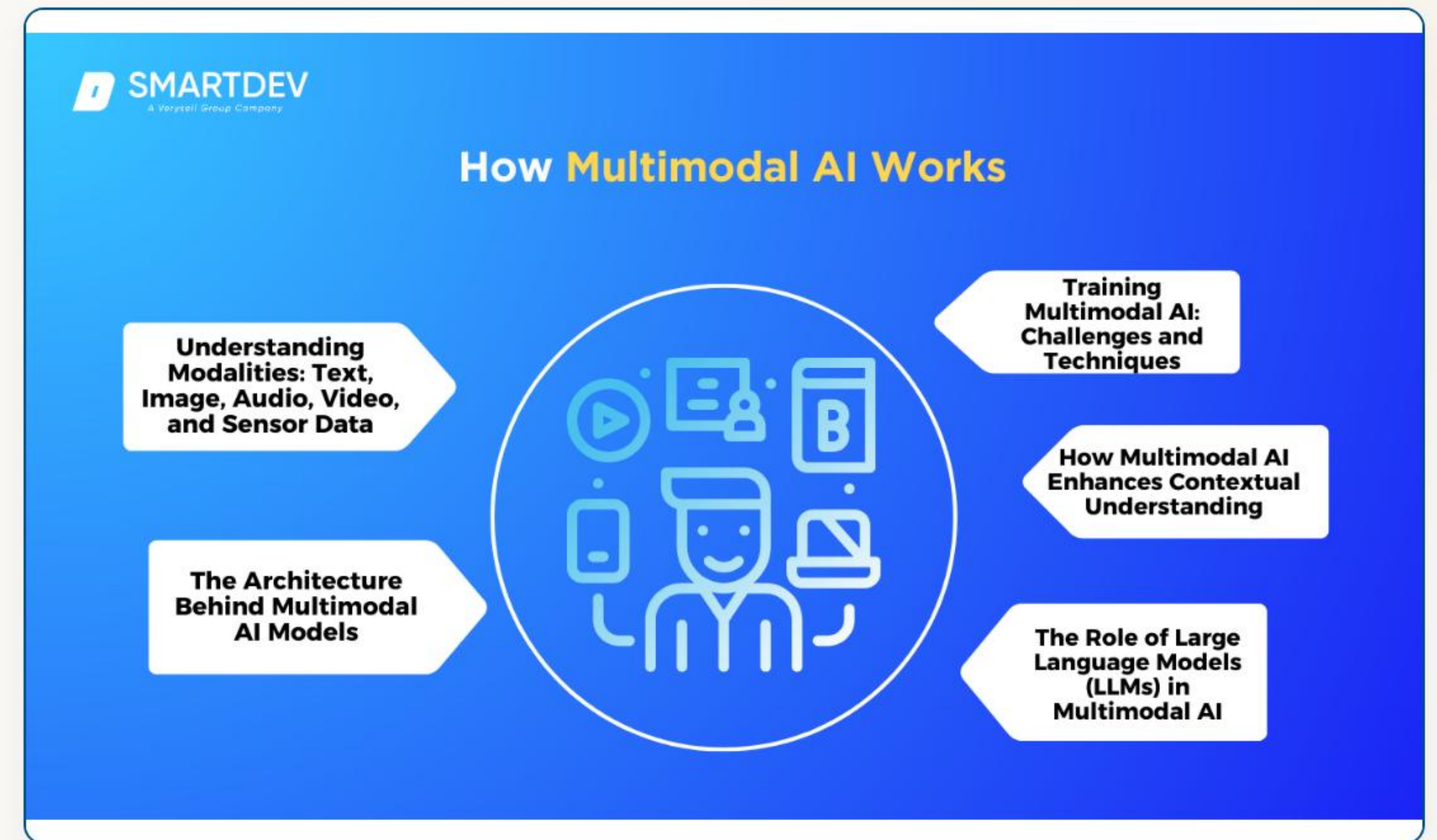
Methodological Steps:

- Top-Down Mind Maps:** LLM recursively decomposes the target question into sub-questions $m = (q, t, s)$ until atomic.
- Dual-Level Retrieval:** Extracts local attributes (entities/triples) and global constraints (connected subgraphs).
- Dual-Process Verification:** Evaluator model LLM_{ver} checks answers bottom-up; triggers a selective "re-thinking" loop on failure.

Benchmarks & Gaps:

Datasets: HotpotQA, CWQ, WebQSP, and GRBench (Academic, Amazon, Goodreads, Disease KGs).

⚠ Critical Gap: High verification API cost and recursive latency; verification can self-loop.



M³KG-RAG: Multimodal Grounding

Park et al., CVPR 2026 | [Park_M3KG-RAG_..._paper.pdf](#)

Methodological Steps:

Modality-Wise Indexing: Connects video/audio items directly to graph triplets to map cross-modal signals.

GRASP Framework: GroundingDINO (visual) and TAG (audio) models evaluate the presence score:

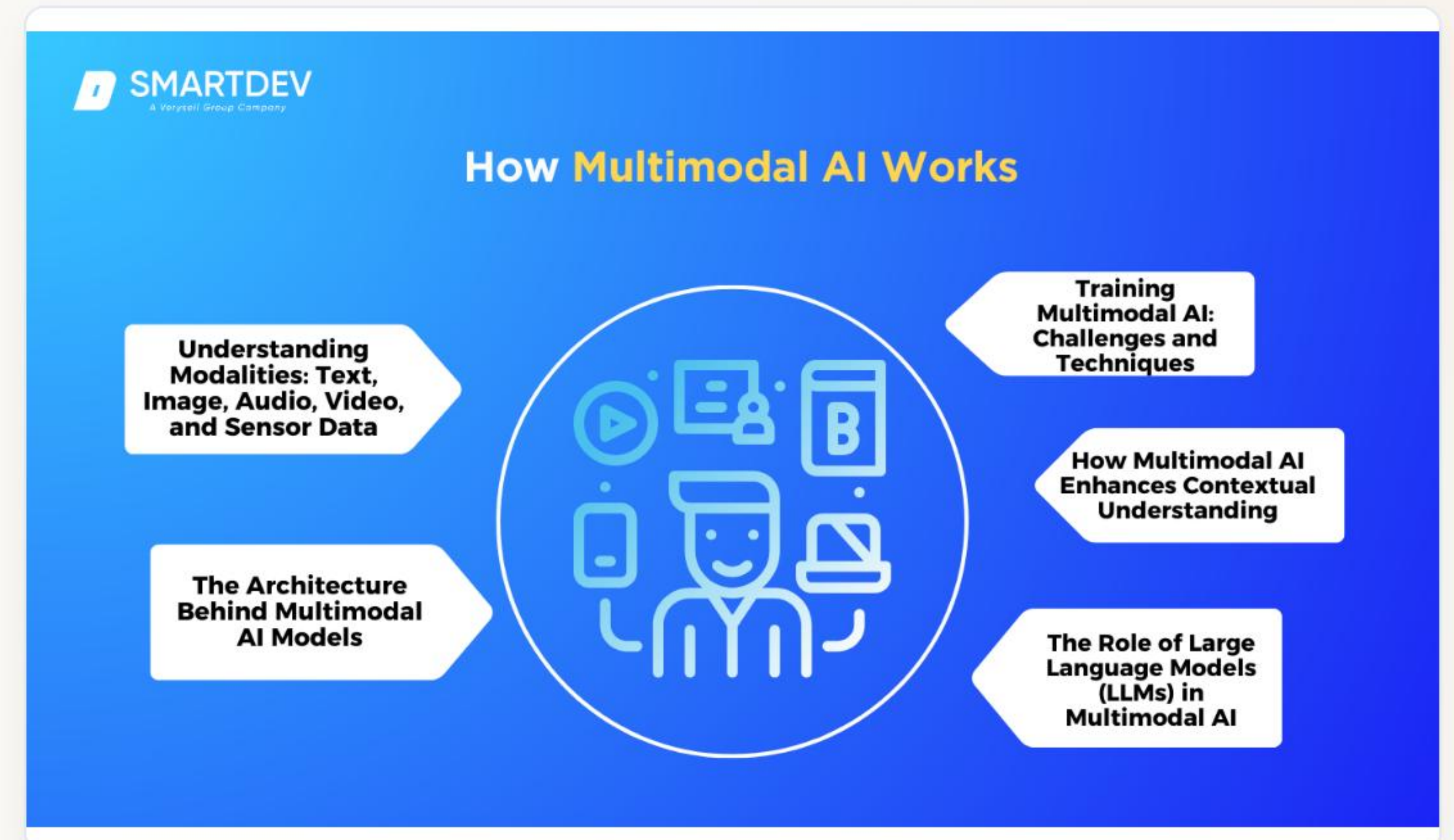
$$s_a(t|q_a) = \Phi_a(\sigma(t); q_a)$$

Selective Pruning: Drops triplets with low relevance or redundant content before generation.

Benchmarks & Gaps:

Datasets: AudioCaps-QA, Video-ChatGPT, VALOR.

⚠ Critical Gap: Audio-visual temporal alignment is heavy, creating video indexing lag.



GraphFlow: Transition Flow Matching

Yu et al., NeurIPS 2025 | *NeurIPS-2025-can-...-Bibtex.bib*

The Flow Matching Paradigm:

Formulates multi-step retrieval as a continuous flow matching problem. Instead of relying on expensive human step supervision, it models search as probability trajectory flows.

-  **Reward Factorization:** Automatically decomposes the final answer's reward to each transition step in the KG.
-  **Policy Navigation:** Uses gradient flows to direct the retrieval policy toward diverse, high-value graph subregions.

Empirical Performance & Gap:

Datasets: STaRK Benchmark (large-scale structured product search databases).

-  STaRK Hit Rate: **+10.0% improvement** over GPT-4o baselines on multi-step reasoning.
-  **Critical Gap:** High latency in dynamic graphs. Changes in graph structure require flow estimator retuning.

KG²RAG: Triplet-MST Skeletons

Zhu et al., 2025 | 22.pdf

Methodological Steps:

 **Graph BFS Expansion:** Performs breadth-first searches around semantic seed chunks to extract adjacent triplets.

 **Maximum Spanning Tree (MST):** Selects highly informative connections and eliminates redundant edges:

$$T_i = \text{MST} (B_i)$$

 **DFS Coherent Splicing:** Reranks and stitches MST paths with DFS into cohesive, readable paragraphs.

Benchmarks & Gaps:

Datasets: HotpotQA, MuSiQue, TriviaQA.

⚠ Critical Gap: Highly dependent on graph density; sparse areas yield empty expansion skeletons.

MST Paragraph-Skeleton Splicing



DFS Tree Traversal → Spliced Coherent Text

SG-RAG: Zero-Shot Subgraphs

Saleh et al., ICNLSP 2024 | [2024.icnlsp-1.45.pdf](#)

Methodological Steps:

> **Cypher Query Mapping:** Bypasses vector search limitations by converting natural queries into structured Cypher statements:

```
MATCH (m:Movie)-[:RELEASE_YEAR]->(y:Year)
WHERE m.name = "Sharon Tate"
RETURN m, y
```

-  **Declarative Subgraph Extraction:** Retrieves explicit factual paths and subgraphs directly from Neo4j.
-  **Ordered Triplet Grouping:** Organizes triplets into subject-relation-object strings to preserve logical sequences.

Benchmarks & Gaps:

Datasets: MetaQA (1, 2, and 3-hop film/actor/director relation paths).

 **Critical Gap:** Highly dependent on translating queries to Cypher perfectly; struggles with schema variance.

Method / Metric	1-Hop Accuracy	2-Hop Accuracy	3-Hop Accuracy
Baseline LLAMA3-8B	24.0%	13.0%	17.0%
RAG (Wikipedia Docs)	42.0%	27.0%	19.0%
SG-RAG (Ours)	90.0%	73.0%	58.0%

HyKGE: Hypothesis Exploration

Jiang et al., ACL 2025 | [2025.acl-long.580.pdf](#)

Methodological Steps:

- ✂ **Hypothesis Output Generation:** Generates highly detailed hypothesis outputs (HO) to expand implicit user intents.
- 🔗 **W2NER Entity Linking:** Word-word relation networks extract clean medical entities and align them with GTE.
- ✂ **HO Fragment Reranking:** Breaks HO into distinct fragments and applies fine-grained rerankers to balance diversity and relevance:

$$RC_{\text{prune}} = \text{Rerank} (RC , \{ C \} ; \text{topK})$$

Benchmarks & Gaps:

Datasets: MMCU-Med, CMB-Exam, CMB-Clin with UMLS & CMeKG.

⚠ **Critical Gap:** Hallucination rate is bound to the quality of the zero-shot generated hypothesis.

HyKGE Clinical Path Construction

User Symptom: "Stomach reflux..." (Sparse Query)

Hypothesis Output (HO): Explores GERD, H2 blockers, Antacids.

KG Paths retrieved: GERD → treated by → Calcium Carbonate.

Synthesized Gaps in Structured RAG



Parsing Bottlenecks

Models like **D-RAG**, **HyKGE**, and **SG-RAG** assume accurate upstream entity linking. Real-world queries are often ambiguous, leading to retrieval cascading errors.



Latency & Scale

Bottom-up self-verification (**CogGRAG**), transition-based flow matching (**GraphFlow**), and multimodal temporal indexing (**M³KG-RAG**) introduce significant processing delays.



Dynamic Graph Scaling

Most frameworks assume static knowledge graphs. Triplet updates require full re-indexing, which is computationally expensive for real-time applications.

| Novel Research Horizon

RMF-RAG Pipeline Novelty

Continuous Differentiable Geodesic Flows

We project the symbolic knowledge graph into a continuous, differentiable **Riemannian Manifold** using hyperbolic geometry (Poincaré ball D^n).

Continuous geodesic trajectory flow replaces NP-hard discrete subgraph walks, enabling direct backpropagation of generative loss through unified vector-symbolic geometry:

$$d(u, v) = \cosh^{-1} 1 + 2 \frac{\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)}$$

Continuous Poincare Disk Hyperbolic manifold projection

Strategic RAG Research Timeline

Mapping the evolutionary trajectory of Knowledge Graph QA systems:

2025 - Trajectories

Differentiable Gumbel-Softmax relaxations and MST skeletons (**D-RAG, KG²RAG, HyKGE**).

2027+ - Riemannian

Continuous neuro-symbolic geodesic flows replacing decoupled pipelines.

2024 - Subgraphs

Declarative queries (Cypher) and zero-shot schema mapping pipelines (**SG-RAG**).

2026 - Cognition

Dual-process verifications, goaltrees, and audio-visual grounding (**CogGRAG, M³KG-RAG, Yano et al.**).

Image Sources



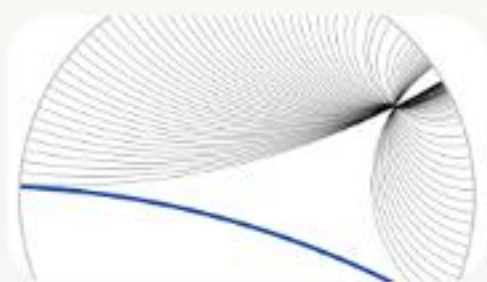
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